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A comprehensive approach to Queue Waiting Time Prediction using Tree-Based Ensembles with Data Balancing and Explainable AI

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Abstract
Queuing up for a service is sometimes an inevitable experience. The inefficiencies brought on by extended waiting times can be considerably decreased by precise waiting time prediction. Accurate prediction can substantially improve consumer satisfaction by reducing uncertainty. It is possible to introduce a robust approach to the prediction of waiting times based on previous queuing data and artificial intelligence (AI) algorithms. This paper contributes to the field by offering a robust approach to waiting time prediction and suggests potential directions for further research. The investigation leverages ensemble tree-based methods along with one statistical model, supplemented by various data pre-processing techniques for regression analysis to forecast precise waiting times. The following regression models have been used to assess the performance: Random Forest (RF), Extra Trees (ET), Gradient Boosting (GBR), Histogram-Based Gradient Boosting (HGBR), Voting (VR) and Ridge Regression. Among these, the ET Regressor demonstrates superior performance. Principal Component Analysis (PCA), t-distributed Stochastic Neighbor Embedding (t-SNE), and Autoencoders have been evaluated to compare the effectiveness of different dimensionality reduction techniques. Furthermore, the challenge of data imbalance in classification tasks has also been addressed here using the Synthetic Minority Oversampling Technique (SMOTE). This process impressively enhances classification accuracy, especially for minority classes. Transparency and trustworthiness in the predictive system have been ensured through the use of Explainable Artificial Intelligence (XAI) techniques, which help interpret the decision-making processes of the models.

Keywords Queue Waiting Time Prediction, Ensemble Techniques, SMOTE, Regression Analysis, XAI, LIME, SHAP

1 Introduction

Managing a queue in a First-come First-served (FCFS) manner is the most straightforward approach. This strategy of processing requests in the exact order of their arrival is particularly effective in scenarios where all clients are treated as equals, ensuring a fair

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and impartial service [1]. In our study, we have utilized a dataset from a banking service where the queuing process was implemented using the same FCFS approach [2].

In any service, the waiting time in a queue plays a big role in client satisfaction. In real life scenarios, waiting in a queue is often unavoidable. However, probable waiting time forecast before standing in a queue can greatly lessen the discontentment that comes with waiting for a lengthy period. So far, numerous approaches have been investigated in the field of waiting time prediction to enhance the precision and effectiveness of service-oriented systems. A prominent method was put in 1999 where customer waiting time was statistically measured based on several parameters like queue length, server count and average service rates [3]. In the meantime, improvements in accuracy have been obtained with an improved version of the queue theory based predictor that integrates the renege rate into the prediction model [4]. Later on, researchers presented a history-based predictor that relies on previous clients' waiting times [5, 6].

Waiting time prediction has also been approached using several advanced techniques like quantile regression and regression splines. Regression splines were used in multi-skill call centre scenarios with waiting period and server count as inputs [7]. Another study investigated quantile regression, concentrating on particular response time percentiles to forecast wait times, especially in hospital emergencies [8]. Such a prediction with Artificial Neural Networks (ANN) has also been proven to be a powerful tool due to its complex modelling and parameter optimization [7]. However, a key limitation of ANNs, especially deep neural networks, is their difficulty with generalization. Though the performance of ANN is quite satisfactory, they are constrained by their black-box nature and lack of hierarchical perception. Training ANNs, particularly deep learning models, is often time-consuming and requires significant computational resources. The process is also energy-intensive, raising sustainability concerns [9, 10].

One noteworthy study used Support Vector Regression (SVR) in the analysis of a queue dataset consisting of real-life data from multiple banks [11]. The study predicted waiting times for individuals and compared the performance of SVR with other machine learning techniques such as K-Nearest Neighbors (KNN) and K-Means Clustering. Though the predictions with SVR are quite impressive compared to different methods, we need to be aware that SVR can distort the ε-tube and lead to overfitting which can result in poor generalization on unseen data [12]. Besides, SVR's computational complexity is an issue with large datasets, as training involves solving a quadratic programming problem [13].

Here, we investigate waiting time prediction on an existing set of data in Bank Teller Lines. A research has already been found on existing dataset to utilize several models like Queuing Theory (QT), Deep Learning (DL), Gradient Boosting Machine (GBM), and Random Forest (RF) [2]. Results revealed that the GBM model has the highest accuracy (97%) and F1-measure (75%). However, their study focused solely on time overflow prediction, which determines whether a client waited for more than 30 minutes. The current study takes a novel approach to the analysis of possible waiting periods for a client by including numerical forecasting in the prediction process. Such a method utilizes regression analysis which is typically used for predicting the exact value of a target variable. We opt for ensemble tree algorithms because conventional regression methods, such as linear regression, have limitations in capturing non-linear relationships in the data [14]. Since we aim to predict waiting time, we feel more appropriate to forecast

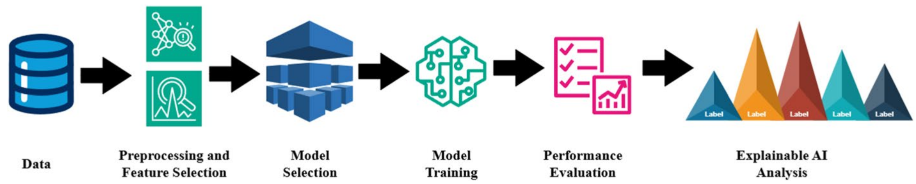

Fig. 1 Flowchart of waiting time prediction process

Table 1 Dataset feature description

Feature Name	Description
branch_id	Unique IDs of respective branches
date	Date of service
queue_id	Unique ID of each queue
sequence	Serial number of the client in the corresponding queue
status	Whether service is finished or reneged
entity	Whether a client is a person or a company
priority	TRUE if the client has any priorities; otherwise, FALSE.
arrival_time	Joining time in the queue
start_time	Service start time
end_time	Service end time
client_id	Unique ID number of the clients
cashier	Unique ID number of the bank personnel who served
service_type	Services that clients received, categorized by A, B, C, and D
waiting_time	The amount of time required to wait before receiving a service

the numerical value of time rather than a discrete output with the help of classification algorithms.

To identify the optimal feature sets, we employ feature selection techniques with forward selection and backward elimination. Concurrently, using the same dataset, we explore a classification approach by categorizing the entire waiting time range into several groups. To rectify the underlying imbalance in class, the study utilizes the SMOTE [15]. Then we evaluate the performance of several classifier algorithms on this balanced dataset. Finally, LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations) are used to introduce explainability of the model performance.

2 Waiting time prediction

This study employs a machine learning approach that are effective tools for analysing and predicting purpose using complex datasets [15]. As shown in Fig. 1, the process begins with the collection of the dataset. Following this, several steps of data preprocessing and feature selection are performed. The refined dataset is then fed into some selected models, which are trained using the dataset. Subsequently, the performance has been evaluated using various metrics and XAI techniques have been employed.

2.1 Dataset

The dataset for this study has been taken from archived teller logs of a Brazilian bank [2]. The logs cover 1 month, from August 1st, 2016, to August 31st, 2016, and contain a total of 12,708 data points. Each data point consists of 14 features, along with a target variable named *waiting_time*, as shown in Table 1. Detailed explanations for all the columns are not available, which makes it challenging to properly understand the statistics. There is a feature column called *service_type* blending four categories: A, B, C, and D. However, the definition of A, B, C, and D is not available in the dataset. Despite the lack of complete information, this dataset aids in our analysis of customer service trends.

2.2 Data preprocessing

Raw data often contain irregularities like missing values, noise, inconsistencies, and redundancies, which can affect the model's performance [16]. Besides, preprocessing of categorical data is essential, as most machine learning models only accept numerical variables. Here, the data pre-processing has been performed with following operations.

i. *Null cancellation*: the *client_id* column has been discarded due to the presence of null values. Presence of null values can introduce bias, complicate pre-processing, and hence significantly affect model interpretation and performance [17].
ii. *Time conversion*: All time-related features have been converted to minutes because it provides a good balance of detail and simplicity compared to seconds or hours when measuring time-related features [18].
iii. *Label encoding*: Label encoding has been applied to categorical features. It is the process of assigning integer representations (like '0', '1' etc.) to all distinct values within a categorical variable [19].
iv. *Addition of new feature*: A feature called *queue_length* has been introduced, representing the number of clients waiting for service when a new client enters the queue as depicted in Fig. 2. The creation of new features or transformation of existing ones allows models to be provided with more pertinent information, enabling them to more effectively capture underlying patterns within the data [20].

2.3 Feature selection

The dataset used in the study contains many features, but not all are crucial for the prediction of waiting time for a client. Removing unnecessary and redundant variables can reduce the amount of features. Here, we employ stepwise feature selection methods, utilizing both forward selection and backward elimination, to refine the model by systematically adding or subtracting features. The stepwise regression approach enables the identification of the most relevant features while avoiding the computational burden of an exhaustive search across all possible feature combinations [21]. In forward selection, the variables are added in succession. It begins by selecting the variable with the highest correlation with the response scores. At each step, the variable that produces the largest

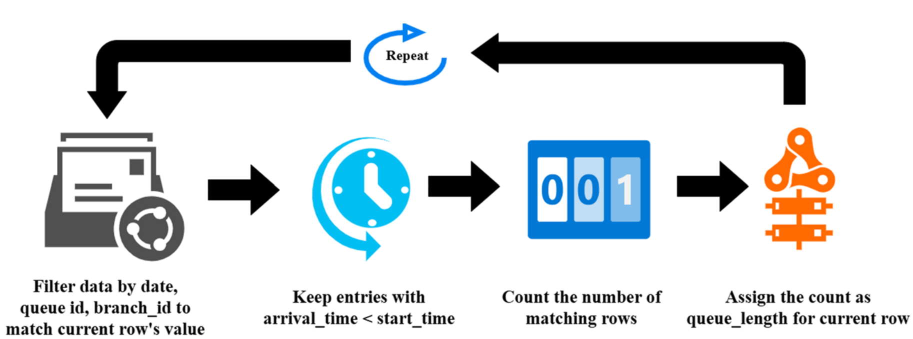

Fig. 2 Queue length calculation process

correlation is attached to the model. Until all variables are incorporated or a threshold is met, the procedure continues to run [22]. On the contrary, backward selection subsequently eliminates variables one at a time, whereas all the predictor variables are previously included. At each step, the variable that gives the lowest score is removed from the model. Until only one variable is left or a predetermined ending criterion is satisfied, this procedure keeps going [22].

2.4 Dimensionality reduction

The mitigation of the undesirable features of high-dimensional spaces is the reason that dimensionality reduction is useful in numerous fields like data compression, noise reduction etc [23–25]. In the present case, we have employed PCA, t-SNE and Autoencoder, which are popular dimensionality reduction techniques used to remove extraneous noise and data while retaining a significant amount of information [26–28]. Following the acquisition by PCA analysis, a stepwise feature selection procedure (discussed in subsection 2.3) has been employed to figure out the optimal set. Such tuning involves the application of both forward selection and backward elimination methods. t-SNE has been used to project the high-dimensional feature space into a 2D representation, effectively preserving local structure and revealing clusters in the data. The resulting t-SNE components have then been used as inputs to ensemble regressors to evaluate the predictive performance of the reduced-dimensional data. This approach highlights the technique's utility not only for visualization but also for assessing the retention of predictive information after dimensionality reduction. Autoencoders, a type of neural network used for non-linear dimensionality reduction, have been employed to learn compressed feature representations by minimizing reconstruction error. The outputs from the encoded layer have subsequently been used as input features for ensemble regressors to evaluate the predictive performance of the reduced-dimensional data.

2.5 Model selection

Previously, a few studies have approached waiting time prediction as a classification task, as their prediction models yielded discrete outputs [2, 29]. However, our study aims to predict waiting time as a continuous numerical output, treating it as a regression task. Since this is a regression problem, ensemble methods are chosen for this task [30]. Ensemble learning initially extracts a set of features through diverse transformations. Using these features, multiple learning algorithms are applied to generate preliminary predictions. The ensemble then combines the information from these predictions, employing adaptive voting schemes to facilitate knowledge discovery and improve overall predictive performance [31, 32]. Both theoretically and empirically, ensemble methods have been demonstrated to outperform single predictors across a wide range of tasks [33]. To start with, RF combines the concepts of bagging and random subspace [34]. It's a powerful predictor due to its ability to avoid overfitting. By introducing appropriate randomness, it excels at classification and regression tasks [35]. ET is introduced as a computationally efficient and highly randomized extension of the random forest algorithm [36, 37]. Lastly, the GBR minimizes a specific loss function with weight distribution adjustments, resulting in the sequential growth of trees. By using forward stage-wise modelling and averaging, it lessens variance and model bias, respectively [38]. HGBR is another histogram-based gradient boosting method that delivers quicker training and less memory consumption [39]. As part of an ensemble approach, the VR creates many models and aggregates their forecasts to provide more accurate predictions [34]. In contrast to bagging, where base models are trained on different subsets of the data, VR involves training each base model on the entire dataset. The final prediction is derived from the combined outputs of all the models [40]. For better performance benchmarking, Ridge Regression, a traditional statistical model is selected. It is a type of linear regression that includes a regularization term to prevent overfitting, especially when dealing with multicollinearity [41]. It modifies ordinary least square regression by adding a penalty proportional to the sum of squared coefficients, which shrinks coefficients toward zero without eliminating them. The regularization strength, controlled by lambda (λ), balances model fit and generalization [42].

2.6 Model training

In this study, the training process involves a systematic approach through splitting the dataset, ensuring reproducibility, and leveraging the capabilities of the scikit-learn library [43]. The dataset is divided into training and test sets using an 80:20 split ratio. This indicates that 80% of the data is allocated for the model training and 20% is left reserved for the performance evaluation. To maintain consistency and reproducibility, a random state of 42 has been employed ensuring that the data-splitting process produces the same results each time when the code is executed.

2.7 XAI analysis

To ensure interpretability in machine learning models used for waiting time prediction, it is essential to understand how individual features contribute to the models' decisions. This study presents a comprehensive analysis of model interpretability using two widely adopted explanation techniques: LIME and SHAP. LIME provides localized, instance-specific explanations by identifying which features most influence a single prediction, offering insights into the directional impact of each input [44]. SHAP, on the other hand, delivers a global perspective by quantifying each feature's contribution across the entire dataset, thereby highlighting consistent patterns of influence [45].

3 Evaluation metrics

Evaluation metrics are numerical measurements that evaluate an algorithm, model, or system's quality and performance. They are essential for figuring out how a model performs on a particular task and for drawing accurate conclusions from performance.

R^2 Score (Coefficient of Determination): Based on the independent variables, the variance of the dependent variable can be predicted and the coefficient of determination can be represented as follows

$$R^2 = 1 - \frac{\sum_{i=1}^{m} (X_i - Y_i)^2}{\sum_{i=1}^{m} \left(\bar{Y} - Y_i\right)^2} \tag{1}$$

here, X_i indicates the probable i^{th} value, and Y_i denotes the true i^{th} value, where $\bar{Y}$ represents the mean of Y_i values.

$$\bar{Y} = \frac{1}{m} \sum {}_{i=1}^{m} Y_i \tag{2}$$

The coefficient of determination assumes values between the worst possible value -∞ to the best possible value + 1, depending on the mutual relationship between the ground truth and the prediction model. R-squared offers an indication of the goodness of fit, assessing the extent to which the observed values align with the predicted values in the regression model [46].

Mean Squared Error (MSE):

The squared difference between the actual and expected values is quantified by MSE. Essentially, it measures the proximity of the line of best fit to the dataset. As a positive value, it is obtained by squaring the differences to eliminate negative signs. A smaller MSE indicates a more accurate prediction, as it signifies a closer alignment between the predicted and actual values [47].

$$MSE = \frac{1}{m} \sum {}_{i=1}^{m} (X_i - Y_i)^2 \tag{3}$$

Root Mean Squared Error (RMSE):

RMSE is derived as the square root of the mean of the squares of all the errors. The standard deviation corresponds to what RMSE stands for. Similar to MSE, RMSE assesses the proximity of the line of best fit to the dataset, providing insight into the accuracy of the prediction [47].

$$RMSE = \sqrt{\frac{1}{m} \sum {}_{i=1}^{m} (X_i - Y_i)^2} \tag{4}$$

Mean Absolute Error (MAE):

The mean of the absolute differences between the predicted and actual values is designated as MAE. Unlike MSE, which computes the squared differences, MAE measures the absolute distinctions [47].

$$MAE = \frac{1}{m} \sum {}_{i=1}^{m} |X_i - Y_i| \tag{5}$$

Accuracy:

The accuracy metric calculates the proportion of accurate predictions to the number of examined instances [48].

$$Accuracy = \frac{tp + tn}{tp + fp + tn + fn} \tag{6}$$

here, tp and tn signifies the number of accurately classified positive and negative instances respectively. Concurrently, fp and fn indicate the count of misclassified positive and negative events, respectively.

F1 Score:

The statistic that strikes a compromise between precision and recall is the F1 Score where precision detects the share of accurately identified objects among the predicted objects and recall signifies the percentage among the actual objects. F1 Score is beneficial in analyzing imbalanced dataset where one class may remain more frequent than the others. It ranges from 0 to 1 representing the poor performance to perfect precision and recall respectively [49].

$$F1 = \frac{2tp}{2tp + fp + fn} = 2 \times \frac{precision \times recall}{precision + recall} \tag{7}$$

Confidence Interval (CI):

A confidence interval (CI) is a range derived from sample data that is expected to contain the true value of a population parameter with a given level of confidence [50]. For example, a 95% CI means that if the same study or sampling process is repeated 100 times, approximately 95 of the resulting intervals are expected to contain the true population parameter. It can be represented mathematically as follows [51]

$$CI = \bar{x} \pm z \times \frac{s}{\sqrt{n}} \tag{8}$$

where, $\bar{x}$ = sample mean, z = critical value from the standard normal distribution corresponding to the desired confidence level, s = sample standard deviation and n = sample size.

Statistical Test:

To evaluate statistical significance, a paired t-test is used to compare the performance of each model against the best-performing model based on their cross-validated R^2 scores. This test assesses the null hypothesis that the mean difference in R^2 scores between two models is zero. A low p-value (typically less than 0.05) indicates that the difference in performance is statistically significant and unlikely to have occurred by random chance. Conversely, a high p-value suggests that the models have similar performance in terms of R^2 scores.

4 Results and discussion

This section evaluates the overall performance of different models based on the performance metrics, along with a comparison of various data preprocessing techniques. Additionally, the outcomes of the classification approach and XAI have been assessed.

4.1 Performance

Here, we evaluate the effectiveness of feature selection methods, dimensionality reduction, and multiple regression models to establish the most accurate approach for predicting waiting times. This analysis underscores the importance of key features and contrasts the behaviour of different machine learning models.

Fig. 3 presents the top-scoring feature sets identified by two feature selection methods, using two different models. In both forward selection and backward elimination, the ET regressor slightly outperformed the RF regressor in terms of R^2 score. The consistency of four features: *cashier*, *queue_length*, *date*, and *arrival_time* in both forward and backward selection methods underscore their significance. This finding is practical, as the service time is greatly influenced by the efficiency of the service provider, in this case, the *cashier*, which significantly impacts the waiting time [52]. In addition, the *queue_length* directly affects the waiting time, as longer queues naturally lead to increased waiting periods [53]. The *date* is crucial because the demand for services can vary significantly on different days [54], impacting the overall waiting time. Similarly, *arrival_time* is also

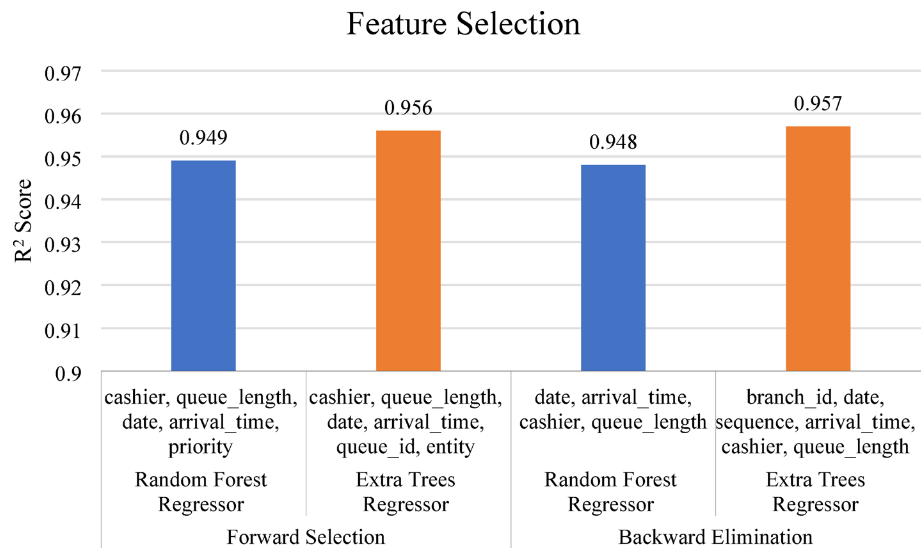

Fig. 3 Top-scoring feature sets based on R^2 Score

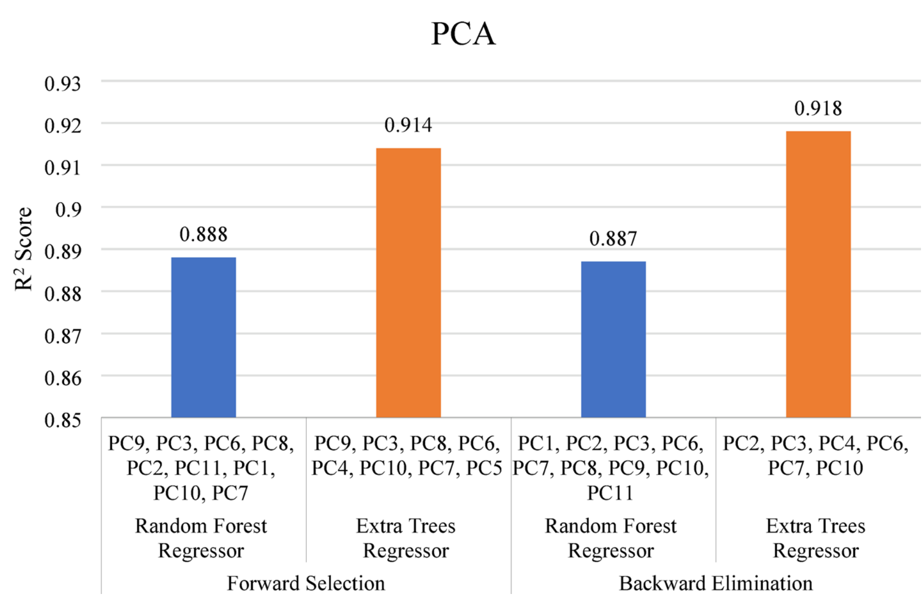

Fig. 4 Top-scoring Principal Component sets based on R^2 score

important because certain hours of the day may experience higher client volume [55], influencing the waiting period. Notably, the feature selection process also helps filter out less interpretable or poorly documented features (e.g., *service_type*), since they do not contribute significantly to model performance. Therefore, the lack of complete dataset documentation has a limited impact on the effectiveness of our analysis.

In Fig. 4, the ET regressor again performed better than the RF regressor in the dimensionality reduction using PCA. However, the widely used dimensionality reduction technique, PCA fails to outperform the original feature sets with unsatisfactory R^2 scores, indicating that dimensionality reduction is not effective in this case. To elaborate, the primary reason for PCA's underperformance is the loss of key, domain-relevant information during transformation [56, 57]. PCA focuses on maximizing variance without considering the target variable, which can result in the compression or removal of features that are crucial for prediction but contribute less to overall variance [58]. Additionally, ensemble models such as ET and RF naturally handle high-dimensional and potentially correlated features without requiring prior dimensionality reduction [59]. These models rely on hierarchical feature splits and can effectively exploit interactions and nonlinearities that PCA, being a linear technique, cannot capture. As such, PCA may misalign with the model structure, introducing a layer of abstraction that reduces interpretability and impairs performance.

As shown in Fig. 5, t-SNE-based features yielded good predictive performance, with R^2 scores of 0.87 for RF and 0.90 for ET. Autoencoder-derived features also performed reasonably, with R^2 scores of 0.79 and 0.83 for RF and ET, respectively. Despite yielding reasonably good results, these dimensionality reduction techniques still fall short compared to the original feature set. This performance gap can be attributed to the inherent trade-off involved in dimensionality reduction. Techniques like t-SNE and Autoencoders are designed to compress high-dimensional data while preserving key structures or patterns, but in doing so, they may discard subtle but relevant variations that contribute to model accuracy [60, 61]. t-SNE focuses on preserving local neighborhood relationships. It does not maintain global structure or variance, which are often important for regression tasks [62]. Autoencoders minimize reconstruction error but are not optimized for prediction [63]. As a result, both methods may lose task-specific information needed for high predictive performance. Notably, feature selection methods such as forward selection and backward elimination are instrumental in identifying the most effective feature sets. The feature set that yields the best R^2 score in these methods has been chosen for waiting time prediction. Hence, further performance evaluation has been conducted using the most significant identified features in conjunction with several ensemble tree-based regression algorithms.

The performance metrics for several models are presented in Table 2. ET regressor emerges as the top performer, showcasing the lowest MAE, MSE, and RMSE scores of 3.06 minutes, 29.82 minutes^2 and 5.46 minutes along with the highest R^2 score of 0.946 with a 95% confidence interval of [0.930, 0.962]. RF closely follows with an R^2 score of 0.934 (95% CI [0.916, 0.950]), and its performance difference compared to ET is statistically significant, with a p-value of 0.004. GBR records a lower R^2 score of 0.78 (95%

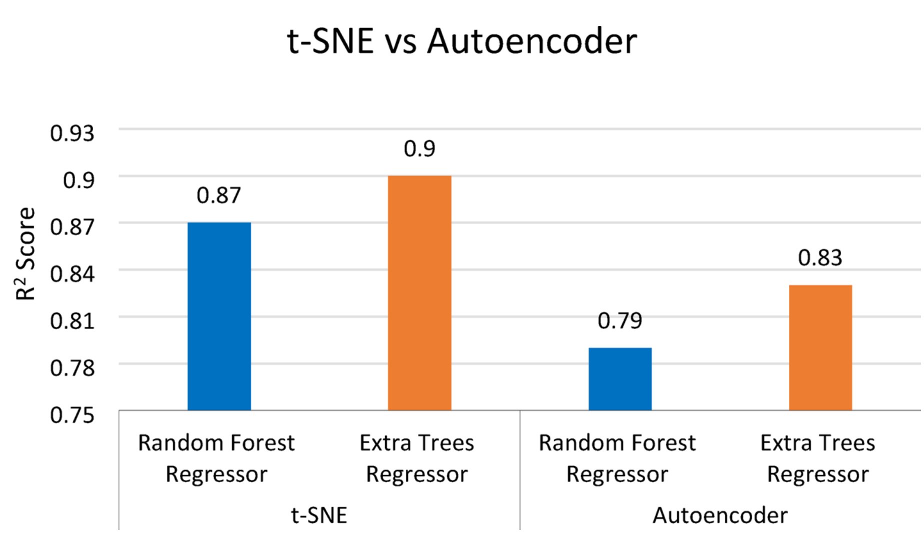

Fig. 5 Performance comparison of t-SNE and autoencoder techniques using two models

Table 2 Model-wise performance metrics

Models	R^2 Score	MAE	MSE	RMSE	95% CI (R^2)	*p*-value (Best vs. Each)
RF	0.934	3.67	34.44	5.86	[0.916, 0.95]	0.004
ET	0.946	3.06	29.82	5.46	[0.93, 0.962]	-
GBR	0.78	7.28	140.29	11.84	[0.76, 0.8]	< 0.001
HGBR	0.89	5.28	62.35	7.89	[0.88, 0.913]	< 0.001
VR	0.88	6.07	95.32	9.76	[0.854, 0.923]	< 0.001
Ridge	0.33	12.42	359.61	18.96	[0.314, 0.355]	<< 0.001

Table 3 Class-wise distribution of waiting times

Class	Class label	Class counts
Very Short (0–5 minutes)	0	5792
Short (6–10 minutes)	1	2212
Normal (11–20 minutes)	2	2092
Long (21–30 minutes)	3	1019
Very Long (> 30 minutes)	4	1593

CI [0.76, 0.8]) among the ensemble models and shows significantly poorer performance with a *p*-value of < 0.001 when compared to ET. Finally, the Ridge Regression model underperforms markedly, with an R^2 score of 0.33 (95% CI [0.314, 0.355]) and a highly significant difference from the best model, supported by a *p*-value of << 0.001. This indicates the limited capability of traditional statistical models to capture the complex relationships in the data compared to ensemble models.

4.2 Classification strategy

For the classification approach, we categorize waiting times into five classes: very short (0–5 minutes), short (6–10 minutes), normal (11–20 minutes), long (21–30 minutes), and very long (> 30 minutes). But the dataset has an issue with class imbalance as shown in Table 3.

The class is imbalanced if not approximately equally represented in the dataset. Our approach to such a problem by under-sampling the classes larger than the minority class reduces the dataset's size, negatively impacting performance. On the contrary, an alternative method called SMOTE unlike traditional over-sampling with replacement, generates synthetic examples to over-sample the minority class, thereby enhancing the balance without reducing the overall dataset size [64].

In the classification technique, the ET classifier outperforms other models. Fig 6, presents the confusion matrix for the ET classifier's results in predicting waiting times using the original dataset. This matrix indicates significantly poor prediction accuracy in classes 1, 2, and 3. Balancing the data in this manner leads to a significant improvement in the classifier model's performance, enhancing its ability to predict waiting times accurately, as displayed in Fig. 7.

4.3 Performance comparison

In this section, we present a comparative performance analysis between our study and that of Mourao et al., utilizing the same dataset. We examine both time overflow prediction and regression analysis, highlighting key differences and improvements with our approach.

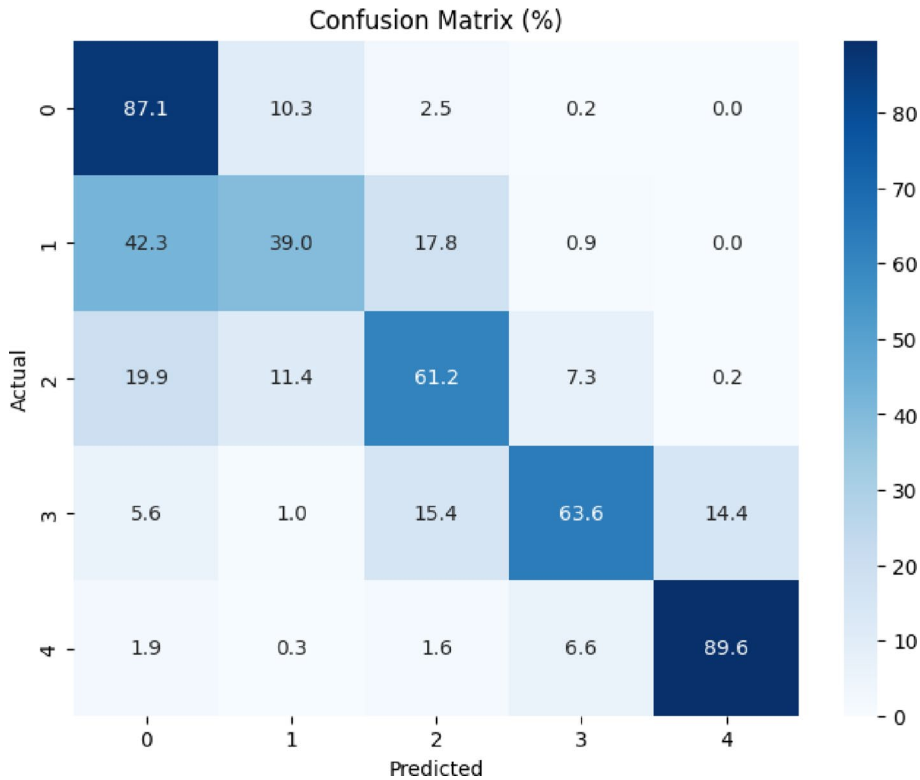

Fig. 6 Confusion matrix of the original dataset

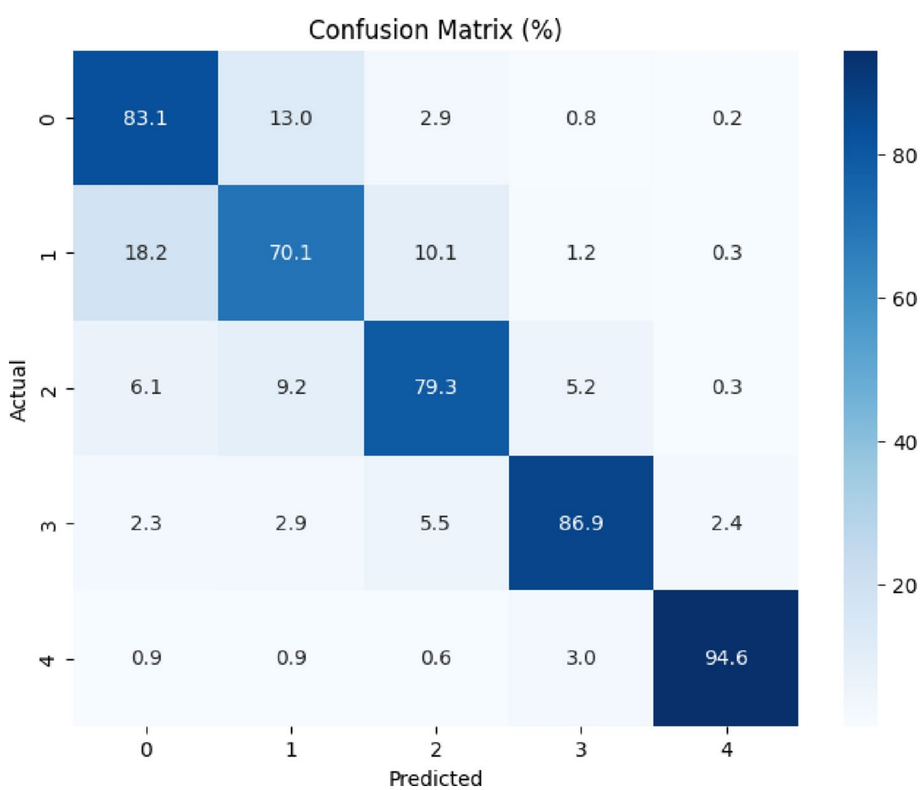

Fig. 7 Confusion matrix after using SMOTE

Table 4 Performance comparison

Methods	Models	Time overflow		Regression	
		Accuracy	F1 Score	R^2 Score	MAE
Mourao et al.	GBM	0.97	0.75	–	–
	RF	0.96	0.68	–	–
Ours	GBM	0.97	0.97	0.78	7.28
	RF	0.97	0.97	0.934	3.67
	ET	0.97	0.97	0.946	3.06

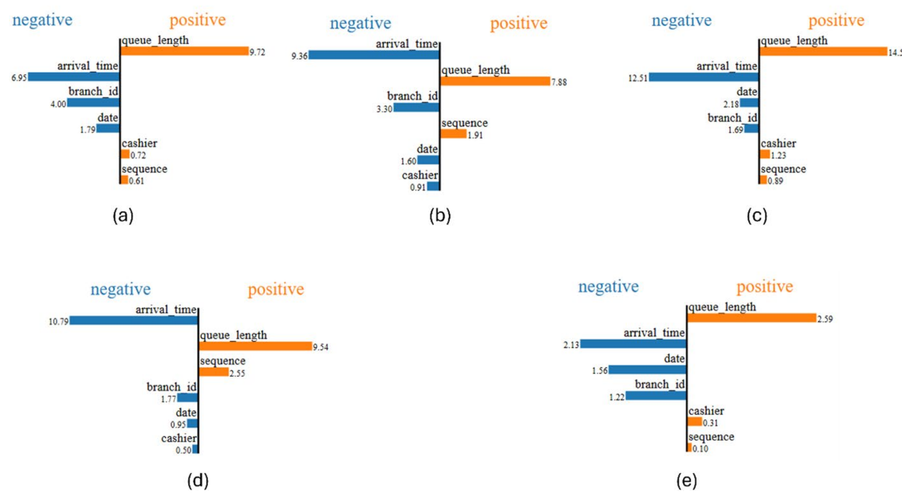

Fig. 8 LIME output showcasing feature contributions across five regression models: **a** Random Forest, **b** Extra Trees, **c** Gradient Boosting, **d** Hist Gradient Boosting, and **e** Voting Regressor

From Table 4, it is evident that in time overflow prediction, Mourao et al. achieved their best result using the GBM model, with an accuracy of 0.97 and an F1 score of 0.75 [2]. In contrast, our study attains the same best accuracy of 0.97 using both RF and GBM models, and an improved F1 score of 0.97 for both models, which is an indication of superior performance in handling imbalanced data and predicting precise waiting times. While the previous study did not explore any regression analysis, the current approach obtains optimal results using the ET regressor model, attaining an R^2 score of 0.946 and an MAE of 3.06 minutes.

4.4 Model explainability

Each subfigure in Fig. 8 visualizes LIME output, the most influential features contributing to a single prediction instance, with features pushing the prediction higher (positive impact) shown in orange, and those pushing it lower (negative impact) shown in blue. The length of the bars represents the magnitude of each feature's contribution. Across all models, *arrival_time* and *queue_length* consistently emerge as dominant features, with *queue_length* contributing positively and *arrival_time* negatively to the predicted waiting time. This behaviour aligns with queuing theory, where longer queues indicate longer delays and earlier arrivals often result in shorter waits. Other features, such as *sequence*, *cashier*, *date*, and *branch_id* show moderate influence depending on the model, indicating that positional order, server allocation, and temporal context also play secondary but meaningful roles in shaping wait time predictions. These LIME-based insights help

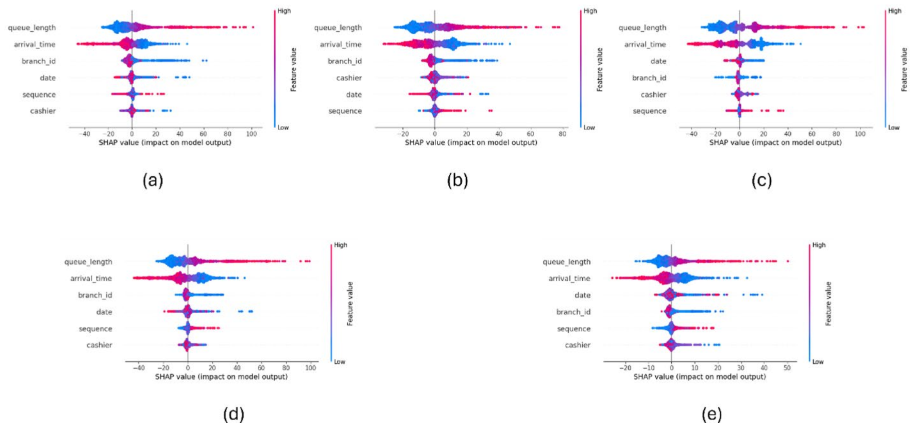

Fig. 9 SHAP-based Global Interpretability using five ensemble regressors: **a** Random Forest, **b** Extra Trees, **c** Gradient Boosting, **d** Hist Gradient Boosting, and **e** Voting Regressor

validate model reasoning and promote transparency in the application of machine learning for operational queue management.

In Fig. 9 each subplot, features are ranked by their overall importance, with horizontal dispersion indicating the magnitude of SHAP values (impact on model output), and color representing the feature value (red = high, blue = low). Consistently across all models, *queue_length* and *arrival_time* are the most influential features affecting waiting time predictions. High values of *queue_length* typically increase the predicted wait time, whereas higher *arrival_time* values (i.e., arriving later) tend to reduce it, as reflected by the gradient of SHAP values. Secondary features such as *cashier*, *date*, *branch_id*, and *sequence* show moderate but non-negligible influence. These capture nuances like server efficiency, temporal variations (e.g., weekdays vs. weekends), and the order of arrival. SHAP outputs thus provide a robust, model-agnostic explanation of how each feature contributes to predictions, reinforcing the reliability and fairness of the deployed models in queue-based decision systems.

5 Conclusion

This study presents a comprehensive and data-driven approach to accurately predicting waiting times, distinguishing itself from prior work by emphasizing precise regression modeling and interpretability. Ensemble tree-based algorithms have been selected for their superior performance. Through meticulous feature selection, the highest R^2 scores have been found in identified key features: *cashier*, *queue_length*, *date*, and *arrival_time*. Ensemble tree models outperform traditional methods, underscoring their ability to capture non-linear relationships in the data. Dimensionality reduction methods such as PCA, t-SNE, and Autoencoders were assessed, with findings indicating that the original feature sets yield better predictive performance. In addition to regression analysis, a classification approach by segmenting waiting times into distinct groups has also been explored. To address class imbalance, the utilization of SMOTE significantly improves classification accuracy, particularly for minority classes. To enhance transparency, LIME and SHAP have been employed, providing insights into individual predictions and overall model behaviour. While this study demonstrates the efficacy of ensemble methods in predicting waiting times, it is constrained by limitations in dataset documentation and contextual detail. The lack of metadata, collection methodology, and clarity on categorical variables restricts the interpretability and generalizability of the results. Future research should prioritize access to more comprehensive, well-documented datasets and explore advanced feature engineering strategies to further improve model performance and insight.

Author contributions

Author Contribution: Tapodhir Karmakar Taton: Data curation (equal), Formal analysis (equal), Investigation(equal), Writing original draft (equal). Bipin Saha: Formal analysis (equal), Investigation (equal), Writing original draft (equal), Md. Johirul Islam: Supervision (equal), Writing original draft (equal), Writing–review and editing (equal). Shaikh Khaled Mostaque: Conceptualization (equal), Supervision (equal), Writing original draft (equal), Writing–review and editing (equal).

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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The authors declare no competing interests.

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